Vortical Enclosure Dynamics, Vol. 3

From Closure Density to Gravity and Cosmology

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Abstract

Vortical Enclosure Dynamics (VED) proposes formal constructions and physical interpretations beginning from a single axiom: *there is difference*. Vol. 1 introduced the causal log C_{ij} and time-like and spatial candidate constructions. Vol. 2 proposed structural correspondences between closure geometry and Standard-Model organization.

This volume addresses gravity. The central claim is that gravity is not a new force, but a modification of the definition of distance. Incorporating closure density L_{ab} into the path cost supplies a formal program relating a metric fixed-point target, weak-field comparisons, non-equilibrium closure phases, and a global $\sigma > 0$ cosmological regime. The physical identifications and exact recoveries remain proposed rather than established.

No additional structures are introduced beyond the closure framework. The present theory does not yet possess quantitative predictive power; structural organization is the primary goal.

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1 Introduction

Vol. 1 introduced time-like accumulation and network-distance constructions from the causal log C_{ij} [1]; their physical identifications remain proposed. Vol. 2 showed that the gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, particle spectrum, and mass generation can be interpreted as structural consequences of triangular closure [2].

The present volume addresses the remaining question: how does gravitational dynamics arise within the same framework?

The starting point is the distance definition of Vol. 1:

$$d_{ij} = -\log \tilde{S}_{ij}, \qquad \tilde{S}_{ij} = \max_{\gamma: i \to j} \prod_{(ab) \in \gamma} S_{ab}. \tag{1}$$

This definition is purely structural, determined by the most permeable path in the causal log network. It yields a consistent geometry, but does not produce gravity. The reason is that it assumes structural homogeneity: no mechanism biases path selection according to accumulated closure structure.

Gravity, within VED, appears precisely as such a bias. The central claim of this volume is:

Gravity is not a new force. It appears as a modification of the definition of distance.

(2)

Incorporating closure density L_{ab} into the path cost unifies geometry and gravitational effect within a single selection principle. From this modification, the volume proposes an intended weak-field comparison with the Einstein equation, a Schwarzschild-type exterior comparison, a non-equilibrium interpretation of the black-hole interior, and a global $\sigma > 0$ cosmological candidate. These are structural targets rather than exact physical recoveries.

This interpretation is structural and qualitative in nature. No quantitative predictions are derived, and no claims are made about numerical correspondence with observed values. The goal is to exhibit a coherent geometric reading of gravitational physics.

The paper is organized as follows. Section 2 introduces the modified distance definition. Section 3 develops the closure tensor and its relation to the metric. Section 4 compares a metric fixed-point target with the weak-field form of the Einstein equation. Section 5 examines the exterior solution. Section 6 treats the black hole interior as a non-equilibrium phase. Section 7 addresses cosmic expansion. Section 8 synthesizes the three volumes. Section 9 discusses scope and limitations.

2 Where Does Gravity Come From?

2.1 The Distance of Vol.1 and Its Limitation

In Vol. 1, distance was defined purely from the causal log structure:

$$d_{ij} = -\log \tilde{S}_{ij}, \qquad \tilde{S}_{ij} = \max_{\gamma: i \to j} \prod_{(ab) \in \gamma} S_{ab}. \tag{3}$$

This definition selects the path of maximal propagation strength. Distance is therefore determined by the most permeable structural path.

$$\boxed{\text{Distance is determined solely by the structure of } C_{ij}.} \tag{4}$$

This construction supplies a formal network geometry from relational inputs. Its identification with physical geometry remains open. An additional model ingredient is then considered.

$$\boxed{\text{Gravity does not explicitly emerge from this definition.}} \tag{5}$$

The reason is that the distance above assumes structural homogeneity. All regions are treated equally except for variations encoded in propagation strength. No mechanism exists that biases path selection according to accumulated structure. Gravity, however, appears precisely as such a bias.

2.2 What Is Gravity?

In conventional physics, gravity is introduced either as a force acting between masses, or as curvature of spacetime. VED adopts a different starting point.

$$\boxed{\text{Gravity is the phenomenon by which closure density distorts the distance structure.}} \tag{6}$$

This represents a shift in perspective. The conventional approach introduces a new interaction. VED modifies the definition of distance itself.

$$\boxed{\text{Gravity is not added. It is a reinterpretation of existing structure.}} \tag{7}$$

Thus the volume proposes gravitational behavior as an effective bias in the same structural geometry already defined in Vol. 1. This is a candidate physical interpretation, not an established identification.

2.3 Extension of Distance

To incorporate this bias, assign a cost to each edge (a, b):

$$w_{ab} = -\log S_{ab} + \epsilon\, L_{ab}. \tag{8}$$

The first term represents geometric permeability, identical to Vol. 1. The second term introduces a congestion cost due to closure density. L_{ab} is the edge-level closure quantity used in this path cost; it is not identified by shared lettering with Vol. 1's scalar L or with the coarse tensor $L_{\mu\nu}$. Their cross-carrier relation is not fixed in this section. Distance is then defined by minimizing total path cost:

$$\boxed{d_{ij} = \min_{\gamma: i \to j} \sum_{(ab) \in \gamma} w_{ab}.} \tag{9}$$

This is a formal weighted-path construction on the stated network, with admissible paths and the displayed edge costs as inputs. It does not by itself derive a continuum gravitational field or complete any of the open carrier bridges. This expression places the network path cost and a gravity-motivated term in a single formal selection rule; it does not establish physical unification. The overall structure is summarized in Figure 1.

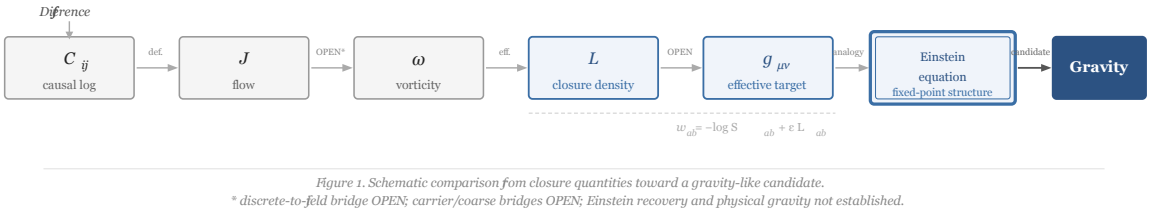


Figure 1: Proposed structural route from closure density to gravity-like behavior. Causal-log, flow, vorticity, closure-density, and metric stages have different formal roles. The Einstein equation is an intended weak-field comparison target, not an exact recovery, and the physical-gravity map remains open.

2.4 What the Unified Path Implies

It is essential that both terms in equation (8) are evaluated along the same path. If propagation strength and closure density were evaluated on different paths, the geometry would split into separate structures.

$$\boxed{\text{Gravity alters path selection itself.}} \tag{10}$$

This mirrors the conceptual shift introduced by general relativity. Einstein replaced force with curvature. VED replaces curvature with path cost. The two descriptions are

compared structurally; equivalence of their physical predictions is not established here.

2.5 Summary

By incorporating closure density into the distance definition, the model offers a candidate effective route to gravity-like behavior within the same structural framework.

$$\boxed{\text{Gravity is not a new force.}} \tag{11}$$

$$\boxed{\text{It appears as a modification of the definition of distance.}} \tag{12}$$

3 Closure Density and Geometry

3.1 The Quantity Called Closure Density

Section 2 introduced L_{ab} as the additional contribution to the path cost:

$$\boxed{L_{ab} = \sum_k \phi_{abk}.} \tag{13}$$

This represents the total contribution of triangular closures containing the edge (a, b). Each triangle contributes to structural congestion, and the sum measures how densely that edge is embedded in closure structure. Intuitively, regions with many triangles are structurally dense; regions with few triangles are structurally sparse.

Remark 1 (Carrier-specific notation)**.** *L_{ab} is an edge quantity in this volume. It is not the scalar $L \sim |\omega|$ of Vol. 1 and is not the tensor $L_{\mu\nu}$ introduced below. No equality among these carriers is implied by the shared letter.*

$$\boxed{\text{Closure density} = \text{density of structure.}} \tag{14}$$

This interpretation follows the Vol. 2 model-local use of triangular closure as the minimal candidate in its stated finite closure class.

3.2 Why It Distorts Distance

The modified edge cost was $w_{ab} = -\log S_{ab} + \epsilon L_{ab}$. The first term favors highly permeable paths; the second penalizes structurally congested regions. Thus, large closure density increases traversal cost.

$$\boxed{\text{Closure density effectively stretches distance.}} \tag{15}$$

This reproduces the qualitative behavior associated with gravity: dense regions become harder to traverse, propagation paths bend around them, and effective geometry is distorted. Unlike general relativity, curvature is not introduced here as a fundamental geometric postulate. The path modification is proposed as an effective candidate for geometry-like distortion.

3.3 Tensorization

Closure density is directional. An edge contributes differently depending on orientation. To describe this, define a coarse-grained tensor over a region U:

$$L_{\mu\nu}(x) = \left\langle L_{ab}\, \hat{e}_{\mu}^{(ab)} \hat{e}_{\nu}^{(ab)} \right\rangle_{U}, \tag{16}$$

where $\hat{e}_{\mu}^{(ab)}$ is the direction vector determined internally from the distance structure. No external embedding space is assumed.

The displayed regional average is the proposed Vol. 3 construction of the tensorial quantity from edge data. It does not identify either quantity with Vol. 1's scalar L, and it does not by itself supply a complete discrete-to-continuum bridge. In particular, sampling and normalization rules, scaling / limiting behavior, uniqueness of the tensorization, and the relation to the scalar L remain open; the displayed construction is a partial bridge.

$$\boxed{L_{\mu\nu} = \text{closure tensor.}} \tag{17}$$

This tensor is intended to encode directional structural congestion within the effective representation and can parameterize anisotropic path-cost changes.

3.4 Relation to the Metric

The modification of distance can be absorbed into an effective metric:

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + \alpha_G L_{\mu\nu}, \tag{18}$$

where $g_{\mu\nu}^{(0)}$ is the proposed effective background metric of Vol. 1, $L_{\mu\nu}$ is the closure tensor, and α_G is a structural coupling constant.

This relation is internal to the proposed continuum description. The bridge from the network distance d_{ij} to $g_{\mu\nu}^{(0)}$ is partial and open: no reconstruction, convergence, normalization, or uniqueness result is supplied here.

$$\boxed{\text{Closure density contributes to the proposed effective metric.}} \tag{19}$$

This is structurally analogous to the statement that matter curves spacetime, but no matter field has been introduced. Whether the effective relation identifies physical geometry remains open.

3.5 G as an Effective Coupling

The coefficient α_G plays the role of gravitational coupling:

$$\boxed{G_{\text{eff}} \sim \alpha_G.} \tag{20}$$

In this representation the constant is introduced as an effective parameter of coarse-grained closure structure; the text does not establish whether a corresponding physical constant is fundamental or derived. At microscopic scales, different particle closures may contribute differently; at macroscopic scales, coarse-graining averages these contributions. This motivates a candidate approximately universal effective coupling; no cross-species convergence derivation is supplied here.

$$\boxed{\text{Universality of } G \text{ is an open coarse-graining target.}} \tag{21}$$

3.6 Summary

Closure density is used as the model source of a candidate gravitational distortion.

$$\boxed{\text{Closure density is a candidate source of gravity-like behavior.}} \tag{22}$$

Under the proposed partial bridge, the modification of distance is represented by an effective metric. Reconstruction and physical interpretation remain open; this statement does not complete the distance-to-metric map.

$$\boxed{\text{Coarse-grained closure geometry is a candidate spacetime description.}} \tag{23}$$

4 The Einstein Equation as a Fixed-Point Structure

4.1 Positioning the Question

So far: distance is derived from C_{ij}; gravity appears as closure density L; the metric takes the form $g_{\mu\nu} = g^{(0)}_{\mu\nu} + \alpha_G L_{\mu\nu}$. Here L names the scalar Vol. 1 closure-density / intensity quantity, while $L_{\mu\nu}$ is the tensorial Vol. 3 quantity. They are not identical by notation alone; the cross-carrier bridge remains separate from the equations used in

this section. The next question is therefore unavoidable:

$$\boxed{\text{What equation does this structure obey?}} \tag{24}$$

In conventional physics, the Einstein equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ is postulated as a fundamental law. Within VED, the situation is reversed.

$$\boxed{\text{The gravitational equation is not assumed.}} \tag{25}$$

Instead, the volume asks whether it can be recovered as an effective target of the self-consistent closure structure.

4.2 Starting Point: The Fixed-Point Equation

Vol. 1 introduced the self-referential structure:

$$C_{ij} = 1 - \exp(-\lambda \rho_i H_{ij}), \qquad \rho_i = \sum_j C_{ij}. \tag{26}$$

This defines a nonlinear fixed-point problem.

$$\boxed{\text{Structure determines itself.}} \tag{27}$$

Geometry, density, and flow are all determined simultaneously by this self-consistency condition. This self-referential structure motivates the proposed comparison with gravitational field equations.

4.3 Continuum Limit and Vorticity

The discrete flow remains $J_{ij} = C_{ij} - C_{ji}$. At the continuum level this section uses a field denoted by J; its connection to J_{ij} is an informal, open bridge because no averaging, scaling, orientation, or vector-field map is supplied. Within that target representation:

$$\omega = \nabla \times J, \qquad L \sim |\omega|. \tag{28}$$

Within this continuum vector-field representation, the first relation defines vorticity and the second is an effective proportional scalar measure. Its magnitude convention and normalization are not fixed, and neither relation is by itself a stability theorem.

$$\boxed{\text{Closure density appears as vorticity of the flow.}} \tag{29}$$

This is an intended connection from microscopic closure structure to macroscopic geometric distortion; the bridge remains open.

4.4 Fixed-Point Condition

In the classical phase established in Vol. 1:

$$\sigma \approx 0, \qquad \nabla \cdot J \approx 0. \tag{30}$$

These conditions correspond to approximate conservation.

$$\boxed{\text{Conservation laws are properties of the phase.}} \tag{31}$$

Under the model interpretation, these conditions identify a candidate stable regime for an effective metric. Stability and physical gravitational geometry are not established by the displayed conditions alone.

4.5 Self-Consistency Condition for the Metric

From Section 3, $g_{\mu\nu} = g^{(0)}_{\mu\nu} + \alpha_G L_{\mu\nu}$. However, $L_{\mu\nu}$ depends on flow, which depends on distance, which depends on the metric itself. Therefore $L_{\mu\nu} = L_{\mu\nu}[g]$, and the metric must satisfy:

$$\boxed{g_{\mu\nu} = g^{(0)}_{\mu\nu} + \alpha_G L_{\mu\nu}[g].} \tag{32}$$

This defines a fixed-point problem for the metric. It is a proposed self-consistency target in the effective continuum representation. The display does not establish a solution, existence, uniqueness, convergence of an iteration, or stability; the construction of $L_{\mu\nu}[g]$ also remains bridge-dependent.

4.6 Linearization and the Form of the Equation

In the weak-gravity limit $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ ($|h_{\mu\nu}| \ll 1$), the following is an intended comparison target for a first-order treatment of equation (32), not a derived expansion here:

$$\boxed{G_{\mu\nu} = 8\pi G_{\text{eff}} L_{\mu\nu}.} \tag{33}$$

$$\boxed{\text{Structurally analogous to the Einstein equation.}} \tag{34}$$

The displayed equation is structurally analogous to the Einstein equation and serves as an intended weak-field target. Precise coefficients, a complete linearization derivation, and nonlinear recovery are not supplied here.

4.7 The Inversion

Conventional understanding: the Einstein equation = a fundamental law.

VED:

$$\boxed{\text{The Einstein equation is an intended weak-field target of the fixed-point structure.}} \tag{35}$$

The proposed comparison is scoped to the classical phase. This does not establish that physical gravity emerges or that the Einstein equation is recovered outside the stated approximation.

4.8 Why General Relativity Works

This interpretation is intended to be compatible with the weak-field success of GR; it does not itself explain or reproduce the full empirical record.

$$\boxed{\text{GR is not incorrect.}} \tag{36}$$

$$\boxed{\text{It is an effective theory of the classical closure phase.}} \tag{37}$$

In this proposed phase description, closure density varies slowly and conservation is treated approximately. A quantitative comparison showing when the effective metric reproduces Einstein dynamics remains open.

5 The Schwarzschild Solution as Exterior Closure

5.1 What This Section Does

Section 4 proposed a fixed-point target and a weak-field structural comparison for gravitational dynamics. The next step is to examine the simplest configuration: static, spherically symmetric, and with localized closure density.

$$\boxed{\text{Closure concentrated at a single point.}} \tag{38}$$

In general relativity, this leads to the Schwarzschild solution. Within VED, the modified-distance model is compared with this exterior form.

5.2 Definition of the Exterior Vacuum

Consider a localized closure distribution $L(x) = M\,\delta(x)$. Outside the localized region ($r > R$), $L(r) = 0$. In this region, closure density vanishes but the metric remains influenced by the boundary conditions imposed by the central structure.

Here the Poisson-type relation is used as an effective weak-field analogy in a continuum scalar-field domain supporting the Laplacian. Its proportionality coefficient,

units, and boundary data belong to that effective specification; it is not a discrete graph derivation or an exact recovery of general relativity. The potential equation becomes:

$$\nabla^2\Psi \propto L = 0 \qquad \Rightarrow \qquad \nabla^2\Psi = 0. \tag{39}$$

For spherical symmetry, the solution is:

$$\boxed{\Psi(r) = -\frac{G_{\text{eff}}M}{r}.} \tag{40}$$

$$\boxed{\text{The Newtonian potential.}} \tag{41}$$

5.3 Form of the Metric

In the weak-gravity limit, the metric takes the form:

$$\boxed{ds^2 \approx -\left(1 - \frac{2GM}{r}\right)dt^2 + \left(1 + \frac{2GM}{r}\right)dr^2 + r^2 d\Omega^2.} \tag{42}$$

This expression corresponds to the first-order expansion of the Schwarzschild metric.

$$\boxed{\text{Consistent with the Schwarzschild solution (weak-gravity limit).}} \tag{43}$$

A full nonlinear recovery would require solving the fixed-point equation beyond linear order and matching the resulting coefficients; that recovery remains open.

5.4 What Is Happening

General relativity interprets this solution as curvature of spacetime. VED interprets it differently.

$$\boxed{\text{Distance cost is modified by closure density.}} \tag{44}$$

Within the effective interpretation, propagation paths and the metric may be compared with bending and time-dilation behavior. Coincidence of observable consequences is not established here.

5.5 VED Interpretation of the Horizon

The Schwarzschild radius is $r_s = 2GM$. At this radius, $g_{tt} \to 0$. VED proposes a path-structure reading of this comparison target, rather than establishing a physical account of geometric breakdown. The following boxed statement names that candidate

interpretation only:

$$\boxed{\text{The horizon is a critical point of path structure.}} \tag{45}$$

In this proposed reading, divergent path cost, frozen propagation, and a critical closure density are candidate features of a horizon phase boundary, as illustrated in Figure 2. Their dynamical realization, stability, and physical identification are not established here.

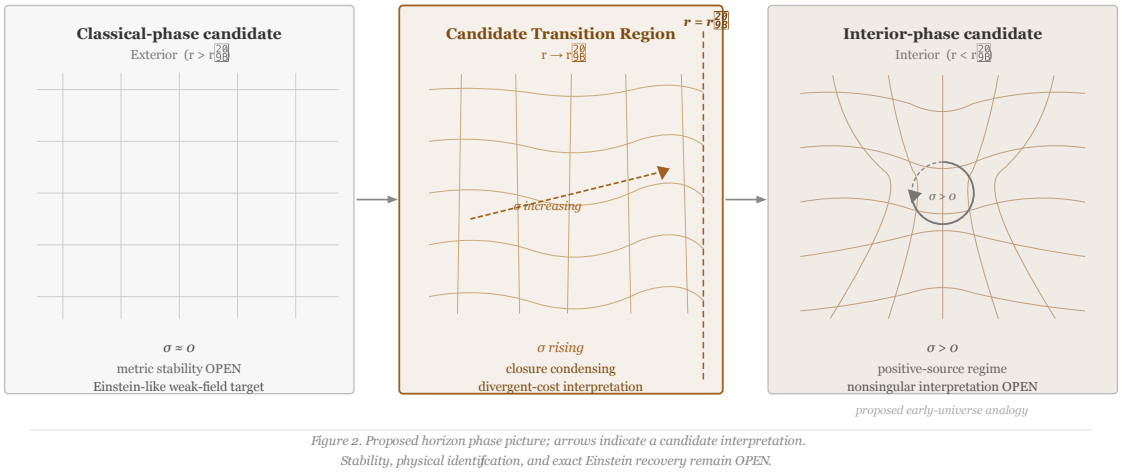


Figure 2. Proposed horizon phase picture; arrows indicate a candidate interpretation. Stability, physical identification, and exact Einstein recovery remain OPEN.

Figure 2: Proposed horizon phase picture, not a demonstrated transition. The exterior ($\sigma \approx 0$), a candidate transition region, and an interior positive-source regime ($\sigma > 0$) are compared schematically. Stability, nonsingularity, and physical identification remain open; the diagram does not establish Einstein recovery.

5.6 Summary

The closure-density model supplies a candidate weak-field exterior comparison.

$$\boxed{\text{A Schwarzschild-type exterior is an intended comparison target.}} \tag{46}$$

No new force is added within this proposed path-cost representation.

$$\boxed{\text{Modified distance provides a candidate gravity-like description.}} \tag{47}$$

6 Black Hole Interior as a Non-Equilibrium Phase

6.1 Revisiting the Meaning of the Horizon

Section 5 proposed comparison of the Schwarzschild radius $r_s = 2GM$ with a critical point of the modified distance structure. At this radius, propagation cost diverges and

the exterior solution ceases to apply.

$$\boxed{\text{The horizon is modeled as a candidate phase boundary.}} \tag{48}$$

This interpretation differs from the conventional geometric breakdown. The model interprets the horizon as a possible transition between two regimes of closure dynamics.

6.2 Exterior vs. Interior

Outside the horizon, the classical phase holds:

$$\sigma \approx 0, \qquad \nabla \cdot J \approx 0, \qquad \partial_t C \approx 0. \tag{49}$$

In this proposed regime, closure structure, approximate conservation, and an effective metric are treated as candidate stable features.

$$\boxed{\text{Exterior} = \text{classical closure phase.}} \tag{50}$$

The model proposes that crossing the horizon changes these conditions.

6.3 Reignition of σ

For the proposed interior regime, the generation term is taken to be nonzero:

$$\boxed{\sigma > 0.} \tag{51}$$

In the model interpretation, log generation resumes and the fixed-point description is no longer imposed: $\partial_t C \neq 0$. This supplies a candidate dynamical-geometry reading, not an established physical interior derivation.

$$\boxed{\text{A time-like accumulation process is modeled as active again.}} \tag{52}$$

The interior is therefore proposed as a non-equilibrium closure regime.

6.4 Collapse of the Fixed Point

In the exterior, $\partial_t C \approx 0$; in the interior, $\partial_t C \neq 0$. The metric is no longer represented by the fixed-point target in this regime.

$$\boxed{\text{The fixed-point description is taken to cease applying.}} \tag{53}$$

An evolving effective-geometry interpretation is proposed rather than derived.

6.5 The Singularity as a Closure Limit

General relativity predicts a singularity at the center. Within VED, the same region is interpreted as the limit of gravitational closure. Using the effective non-closure barrier introduced in Vol. 1,

$$B(L) = -\kappa_B \log\left(1 - \frac{L}{L_{\max}}\right) \to \infty \qquad (L \to L_{\max}). \tag{54}$$

$$\boxed{\text{Perfect closure is not an ordinary attainable state.}} \tag{55}$$

Thus the center is not treated as an ordinary object of infinite density within the VED descriptive window. It is read as an asymptotic closure limit. In the language developed later in Vol. 4, this limit is a local Differential Horizon of the gravitational description: the point where closure attempts to erase the differences required for the description itself.

6.6 Interior Dynamics

Within the proposed $\sigma > 0$ regime, closure structure is modeled as being continuously regenerated.

$$\boxed{\text{The interior is a candidate structural-regeneration regime.}} \tag{56}$$

Logs are generated, closure reorganizes, and geometry evolves. The interior is therefore dynamically active rather than a state of collapse.

6.7 The Inversion

Conventional interpretation: black hole = endpoint.

VED:

$$\boxed{\text{Black hole = entry into a new phase.}} \tag{57}$$

The exterior is a fixed world; the interior is a world under generation.

6.8 Correspondence with the Early Universe

The early-universe and black-hole-interior models both use a $\sigma > 0$ regime.

$$\boxed{\text{The early universe and black-hole interior are proposed as analogous } \sigma > 0 \text{ regimes.}} \tag{58}$$

6.9 Conclusion

$$\boxed{\text{A non-singular interior is a proposed model interpretation.}} \tag{59}$$

$$\boxed{\text{The interior is modeled as a non-equilibrium closure phase.}} \tag{60}$$

Closure evolution and dynamical effective geometry remain candidate physical interpretations.

7 Cosmology: The Global $\sigma > 0$ Phase

7.1 Shift in the Question

Section 6 identified the black hole interior as a local non-equilibrium closure phase characterized by $\sigma > 0$. This raises the natural global question:

$$\boxed{\text{What happens when the } \sigma > 0 \text{ phase extends across the entire universe?}} \tag{61}$$

Within the VED framework, the proposed interpretation is:

$$\boxed{\text{A global } \sigma > 0 \text{ regime is a candidate expansion phase.}} \tag{62}$$

Thus cosmology is modeled as a possible phase of the closure dynamics rather than being derived here as a physical consequence.

7.2 The Fundamental Equation

From Vol. 1, the continuity equation for log density is $\partial_t \rho + \nabla \cdot J = \sigma$. The symbols t and σ in this section retain their Vol. 1 source-local roles as derivative / ordering parameter and continuum log-generation rate. They are not identified here with IFGT's effective external t or its separately constructed generation term σ. Under the homogeneous and isotropic approximation, $\nabla \cdot J \approx 3H\rho$, which yields:

$$\boxed{\dot{\rho} + 3H\rho = \sigma.} \tag{63}$$

The first term represents dilution due to expansion; the second represents generation of structure.

7.3 The Equation in τ-Time

Using the time definition from Vol. 1, $d\tau/dt = \rho$, the equation becomes:

$$\boxed{\frac{d\rho}{d\tau} + 3H_\tau \rho = \sigma(\rho).} \tag{64}$$

The scale factor is defined structurally as $a(\tau) \sim \langle d_{ij}(\tau) \rangle$. As an additional effective ansatz, take $H_\tau \sim \rho$, giving:

$$H^2 \propto \rho, \tag{65}$$

which is structurally analogous to the Friedmann equation.

7.4 Final Form

Substituting $H_\tau \sim \rho$, the evolution equation becomes:

$$\boxed{\frac{d\rho}{d\tau} + 3\gamma\rho^2 = \sigma(\rho).} \tag{66}$$

Here $3\gamma\rho^2$ represents dilution due to expansion and $\sigma(\rho)$ represents generation of logs. The displayed effective model treats cosmological evolution as competition between source and dilution terms.

7.5 The Form of σ

Motivated by the self-referential structure of Vol. 1, we adopt the following phenomenological effective form:

$$\boxed{\sigma(\rho) = \xi\rho\left(1 - \frac{\rho}{\rho_{\max}}\right).} \tag{67}$$

This corresponds to growth at low density and saturation at high density. The logistic form encodes the proposed low-density growth and high-density saturation. It is not established here as the unique or necessary form.

7.6 Fixed Point and de Sitter Phase

The fixed-point solution satisfies $3\gamma\rho_*^2 = \sigma(\rho_*)$, giving:

$$\rho_* = \frac{\xi}{\xi/\rho_{\max} + 3\gamma}. \tag{68}$$

At the fixed point, $H_* = \gamma\rho_* = \text{const}$, and therefore:

$$\boxed{a(\tau) \sim e^{H_*\tau}.} \tag{69}$$

$$\boxed{\text{de Sitter-type accelerated expansion.}} \tag{70}$$

7.7 Origin of the Cosmological Constant

The effective cosmological constant is:

$$\boxed{\Lambda_{\text{eff}} \sim \gamma^2 \rho_*^2.} \tag{71}$$

Within this parameterized effective model, Λ_{eff} is associated with the fixed-point closure scale. Its identification with the observed cosmological constant remains open.

7.8 The Inversion

Conventional cosmology: expansion = initial condition.

VED:

$$\boxed{\text{Cosmic expansion is modeled as phase selection.}} \tag{72}$$

Within the displayed homogeneous model, $\sigma = 0$ and $\sigma > 0$ label source-free and positive-source regimes. They do not by themselves establish necessary and sufficient conditions for physical cosmic expansion.

7.9 Unification with Black Holes

The black-hole-interior and expanding-universe candidate descriptions both use a $\sigma > 0$ regime.

$$\boxed{\text{Black-hole interiors and cosmic expansion are proposed as analogous } \sigma > 0 \text{ regimes.}} \tag{73}$$

7.10 Conclusion

Cosmology is given a candidate global closure-dynamics interpretation.

$$\boxed{\text{Continuous structural generation is a cosmological model proposal.}} \tag{74}$$

$$\boxed{\sigma > 0 \text{ marks a candidate positive-source expansion regime.}} \tag{75}$$

Connecting this formal regime to observed large-scale dynamics requires the open continuum and physical-interpretation bridges.

8 Synthesis: The World as Structure

8.1 The Flow of This Theory

This work began from a single axiom:

$$\boxed{\text{There is difference.}} \tag{76}$$

In the stated representations, difference is associated with gradients, gradients with an effective flow relation, flow with defined vorticity, and vorticity with a candidate enclosure mechanism.

$$\boxed{\text{The model proposes a route from dynamics to structure.}} \tag{77}$$

8.2 The Overall Structure

The following historical compact display summarizes the program's topic order:

$$\boxed{\text{Difference} \rightarrow C_{ij} \rightarrow J \rightarrow \omega \rightarrow L \rightarrow g \rightarrow \text{Einstein equation} \rightarrow \text{Universe}} \tag{78}$$

Quantity	Meaning
C_{ij}	causal log
J	flow of logs
ω	vorticity
L	closure density
g	effective metric
Einstein equation	fixed-point structure
Universe	global $\sigma > 0$ phase

$$\boxed{\text{Everything forms a single structural chain.}} \tag{79}$$

Here "single structural chain" means a connected research program, not one repeated logical relation. The edge roles are:

$C_{ij} \rightarrow J_{ij}$ discrete definition;

$J_{ij} \rightarrow \mathbf{J}(x)$ informal discrete-to-continuum bridge, OPEN;

$\mathbf{J}(x) \rightarrow \omega$ continuum curl definition;

$\omega \rightarrow L$ effective proportional relation;

$L, L_{ab} \rightarrow L_{\mu\nu}$ carrier / coarse bridge, OPEN;

network / closure quantities $\to g_{\mu\nu}$ partial effective endpoint, with reconstruction and physical interpretation OPEN;

$g_{\mu\nu} \to$ **Einstein equation** intended weak-field structural comparison, not exact recovery;

Einstein comparison $\to$ **Universe** proposed cosmological interpretation, not an established physical derivation.

Thus the compact arrows do not constitute a new arrow calculus or establish physical emergence by themselves.

8.3 Correspondence Across Volumes

The three volumes form a connected program whose transitions retain the distinct roles listed above.

Vol. 1 — Generation: causal log C_{ij}; time $d\tau/dt = \rho$; space $d_{ij} = -\log \tilde{S}_{ij}$. Formal time-like and spatial constructions are supplied; their physical interpretations remain proposed.

Vol. 2 — Structure: triangular closure; gauge symmetry $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$; particles, charge, mass. Structure stabilizes.

Vol. 3 — Dynamics: closure quantities enter an effective gravity-like proposal; the metric fixed-point relation is a formal target compared structurally with the weak-field Einstein equation; and the $\sigma > 0$ phase is given a proposed cosmological interpretation. The carrier, continuum, exact-recovery, and observational bridges remain open.

$$\boxed{\text{Generation, structure, and dynamics are compared in one program.}} \tag{80}$$

8.4 Reinterpreting Conservation Laws

Conventional physics treats conservation laws as fundamental; violations appear as exceptions. Within VED, the continuity equation is:

$$\partial_t \rho + \nabla \cdot J = \sigma. \tag{81}$$

Two regimes appear: $\sigma = 0$ implies conservation holds; $\sigma > 0$ denotes a positive source regime in this continuum model; an equivalence with physical generation is not established.

$$\boxed{\text{The source-free balance relation is a model-phase regime.}} \tag{82}$$

This regime classification does not establish whether physical conservation laws are fundamental or emergent.

8.5 Reinterpreting Gravity

Conventional interpretations treat gravity as force or as curvature. VED interprets it differently.

$$\boxed{\text{Modified distance is a candidate gravity-like description.}} \tag{83}$$

$$\boxed{\text{Closure density is a candidate gravity source.}} \tag{84}$$

8.6 Black Holes and the Universe

The black-hole-interior and expanding-universe candidate descriptions both use a $\sigma > 0$ regime.

$$\boxed{\text{Black-hole interiors and cosmic expansion are proposed as analogous } \sigma > 0 \text{ regimes.}} \tag{85}$$

8.7 The Final Inversion

Conventional physics begins by assuming spacetime, fields, and symmetry. VED instead proposes structural constructions and interpretation maps.

$$\boxed{\text{Closure structure supplies the common model input.}} \tag{86}$$

Time-like ordering and network distance are formally constructed. Physical time and space remain proposed interpretations. Particles are represented by candidate closure minima. Forces are flow. The universe is a phase.

8.8 The Final Statement

$$\boxed{\text{The world is not closed.}} \tag{87}$$

$$\boxed{\text{It exists by never fully closing.}} \tag{88}$$

This statement serves both as the starting point and the conclusion of VED.

$$\text{Difference} \rightarrow C_{ij} \rightarrow J \rightarrow \omega \rightarrow L \rightarrow g_{\mu\nu} \rightarrow \text{Einstein equation} \rightarrow \text{Universe} \tag{89}$$

$$\boxed{\begin{aligned} \text{Time-like quantity} &= \text{log accumulation} \\ \text{Spatial candidate} &= \text{network log-distance} \\ \text{Particle-like candidate} &= \text{closure minimum} \\ \text{Gravity-like candidate} &= \text{effective closure-density potential} \\ \text{Cosmological candidate} &= \text{positive-source phase model} \end{aligned}} \tag{90}$$

$$\boxed{\text{The model constructions share } C_{ij} \text{ as an upstream input.}} \tag{91}$$

9 Discussion

9.1 Relation to Vol. 1 and Vol. 2

Vol. 1 introduced time-like and network-spatial constructions from the causal log C_{ij}. Vol. 2 proposed structural comparisons between triangular closure and particle-physics organization. The present volume adds an effective gravity program based on closure density; the physical mappings are not established by these constructions alone.

The key unifying element is the closure energy functional. In Vol. 1, it participated in the proposed metric construction. In Vol. 2, its local minima were used as particle-like candidates. In Vol. 3, its coarse-grained form is proposed as an input to a gravity-like field model. All three volumes therefore share a common geometric foundation.

9.2 Scope and Limitations

The interpretations advanced in this volume are structural and qualitative. Several points of caution are warranted.

First, the identification of $G_{\mu\nu} = 8\pi G_{\text{eff}} L_{\mu\nu}$ with the Einstein equation is structural. No derivation is given of the precise numerical coefficient 8π, nor of the relation between G_{eff} and observed gravitational coupling. A rigorous derivation would require a more developed formalism.

Second, only a weak-gravity Schwarzschild-type comparison is displayed. A full nonlinear recovery from the fixed-point equation has not been demonstrated.

Third, the cosmological equation (66) is structurally analogous to the Friedmann equation, but the precise correspondence between ρ and energy density, and between γ and the equation of state, has not been established.

Fourth, the claim that the black hole interior corresponds to a $\sigma > 0$ phase, while motivated by the non-closure constraint, remains a qualitative assertion. A quantitative treatment would require extending the fixed-point formalism into the non-equilibrium regime.

9.3 Relation to General Relativity

The present framework does not compete with general relativity. GR is accurate in the classical closure phase ($\sigma \approx 0$), and the present work is consistent with this. The distinction arises in other phases: at the black hole interior, at the early universe, and at any regime where the fixed-point structure breaks down.

9.4 Path Forward

Several directions suggest themselves for future development.

The most immediate is a more rigorous treatment of the continuum limit: deriving the closure tensor $L_{\mu\nu}$ explicitly from the discrete network, and showing that the linearized fixed-point equation takes the form $G_{\mu\nu} = 8\pi G_{\text{eff}} L_{\mu\nu}$ without additional assumptions.

A second direction is the cosmological model. The logistic form of $\sigma(\rho)$ is motivated by self-referential selection, but its precise form should be derivable from the fixed-point structure. Connecting the fixed-point density ρ_* to observed cosmological parameters would constitute a non-trivial test of the framework.

A third direction concerns the non-equilibrium phase. The interior dynamics of $\sigma > 0$ regions — including the black hole interior and the early universe — require a treatment beyond the classical fixed-point approximation. This may connect to information-theoretic questions about black hole evaporation and the origin of entropy.

10 Conclusion

This volume has shown that gravitational dynamics can be interpreted as arising from closure geometry. Beginning from the modified distance definition $w_{ab} = -\log S_{ab} + \epsilon L_{ab}$, we have identified:

- modified distance as a candidate gravity-like description (Section 2);
- the closure tensor $L_{\mu\nu}$ as a proposed contribution to metric distortion (Section 3);
- the Einstein equation as an intended weak-field comparison target (Section 4);
- a Schwarzschild-type exterior as a structural comparison target (Section 5);
- the black hole interior as a non-equilibrium closure phase with $\sigma > 0$ (Section 6);
- cosmic expansion as the global $\sigma > 0$ regime, structurally analogous to the Friedmann equation (Section 7).

The following display is the program's compact navigation mnemonic:

$$\text{Difference} \;\to\; C_{ij} \;\to\; J \;\to\; \omega \;\to\; L \;\to\; g_{\mu\nu} \;\to\; \text{Einstein equation} \;\to\; \text{Universe}. \tag{92}$$

Its arrows have distinct roles: $C_{ij} \to J_{ij}$ is a discrete definition; $J_{ij} \to \mathbf{J}(x)$ and the closure-carrier transitions are OPEN bridges; $\mathbf{J} \to \omega$ is a definition; $\omega \to L$ is an effective relation; the metric endpoint is partial and bridge-dependent; and the Einstein and Universe endpoints are structural comparisons or proposed physical interpretations. The mnemonic is therefore not a uniform derivation or generation mechanism.

Taken together with Vol. 1 and Vol. 2, this gives a connected program of formal constructions and proposed interpretations for time, space, matter, and gravity beginning from the axiom *there is difference*. It is not an established physical derivation of those targets. It should not be read as final closure of the framework. Vol. 4 further clarifies why complete closure appears as a descriptive limit rather than as an attainable state.

The interpretation is structural and qualitative in nature, and does not yet constitute a predictive theory. Quantitative correspondence with observation remains a goal for future work.

$$\boxed{\text{The world is not closed. It exists by never fully closing.}} \tag{93}$$

References

[1] Yuhki Endoh. Vortical enclosure dynamics, vol. 1: From difference to space and time, 2026. Independent research draft.

[2] Yuhki Endoh. Vortical enclosure dynamics, vol. 2: From closure geometry to the standard model, 2026. Independent research draft.

A Notation

Symbol	Definition	Source
C_{ij}	causal log matrix	Vol. 1
$\rho_i = \sum_j C_{ij}$	log density	Vol. 1
S_{ab}	symmetric component of C_{ij}	Vol. 1
$\tilde{S}_{ij}$	maximal path propagation strength	Vol. 1
d_{ij}	weighted shortest distance	Vol. 3 §2
$w_{ab} = -\log S_{ab} + \epsilon L_{ab}$	modified edge cost	Vol. 3 §2
$L_{ab} = \sum_k \phi_{abk}$	local closure density	Vol. 2+Vol. 3
$L_{\mu\nu}$	closure tensor (coarse-grained)	Vol. 3 §3
$J_{ij} = C_{ij} - C_{ji}$	log flow (antisymmetric)	Vol. 3
$\omega = \nabla \times J$	vorticity	Vol. 3
σ	Vol. 1-derived continuum log generation rate	Vol. 1+Vol. 3
α_G	gravitational coupling constant	Vol. 3
$G_{\text{eff}} \sim \alpha_G$	effective gravitational constant	Vol. 3 §3
$g^{(0)}_{\mu\nu}$	background metric from Vol. 1	Vol. 1
$g_{\mu\nu} = g^{(0)}_{\mu\nu} + \alpha_G L_{\mu\nu}$	effective metric	Vol. 3 §3
Ψ	gravitational potential	Vol. 3 §4
$r_s = 2G_{\text{eff}}M$	Schwarzschild radius	Vol. 3 §4
$\sigma(\rho)$	effective model of log generation rate	Vol. 3 §6
ρ_*	cosmological fixed-point density	Vol. 3 §6
H_*	de Sitter expansion rate	Vol. 3 §6
$\Lambda_{\text{eff}} \sim \gamma^2 \rho_*^2$	effective cosmological constant	Vol. 3 §6